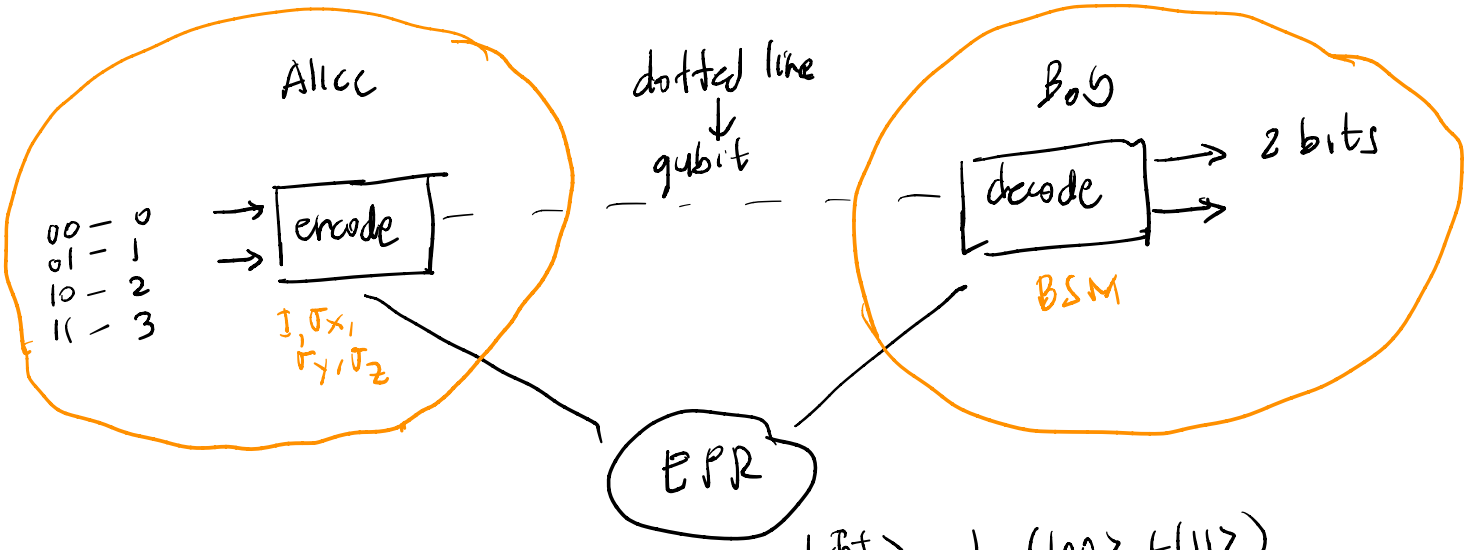
14/8/2026

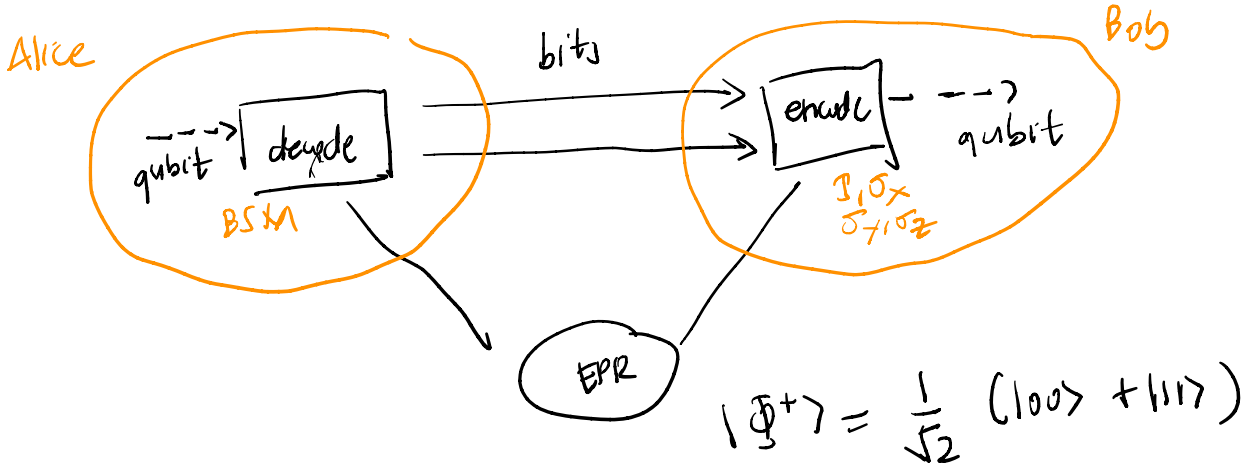
Review:

Dense coding - pre-shared entanglement
B receives 1 qubit from A, extracts 2 bits



$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Teleportation - pre-shared entanglement
B receives 2 bits, 'recreates' 1 qubit

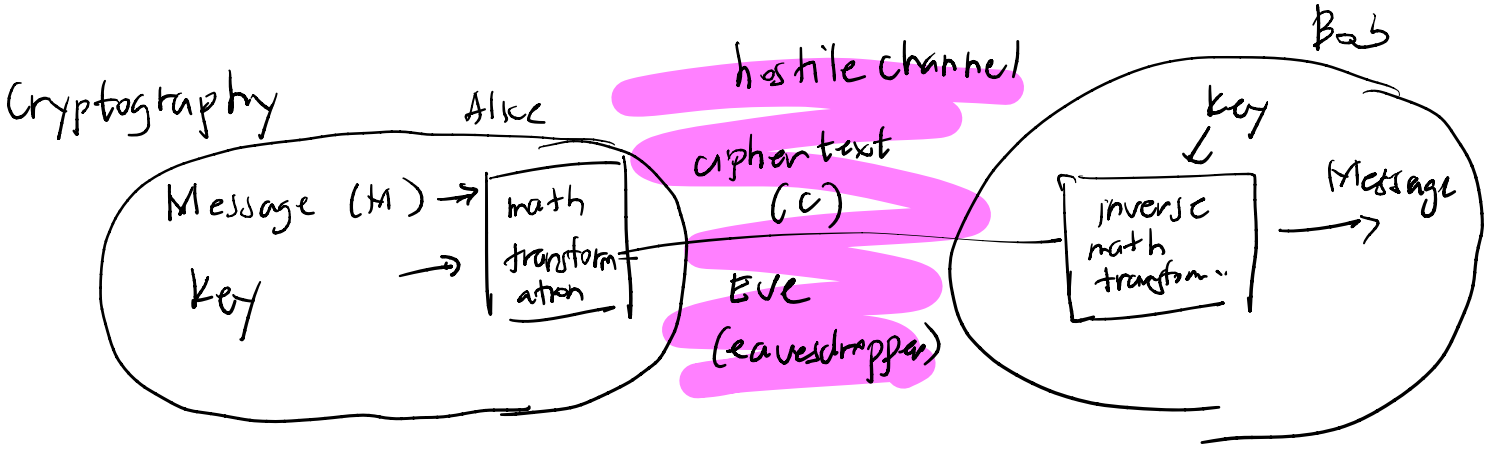


$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Quantum communications — e.g. QKD < BB84
E91

BB84 - Bennett & Brassard - complementarity
- indivisibility of single photon

E91 - Ekert - entanglement



Task of Eve: reconstruct M given C

The message is perfectly secret if C gives Eve no information about M. $\{M_i\}$ = set of possible plain text

$\{C_j\}$ = set of possible ciphertext

Perfect security: $P(M_i|C_j) = P(M_i) \quad \forall\ i, j$.

For perfect secrecy: the number of keys must at least be as great as the number of messages

Vernam cipher $C = M \oplus K$ (modulo 2 addition)

ciphertext

$$P(C_j|M_i) = \frac{1}{2^N} = P(C_j)$$

N is the number of bits

Conditional probability:

$$P(a_i|b_j) = \frac{P(b_j|a_i)\,P(a_i)}{P(b_j)}$$

$$P(a_i, b_j) = P(a_i|b_j)\,P(b_j)$$

$M = c \oplus K \oplus K$ — Why random?

QKD - allows Alice & Bob to share a random bit string

BB84 - (slide)

works because of ① no cloning theorem

$U|0\rangle|i\rangle \rightarrow |0\rangle|0\rangle$

$U|1\rangle|i\rangle \rightarrow |1\rangle|1\rangle$

no machine like this

1982 Wootters, Zurek

Ghirardi (letter to editor)

② complementarity

③ photon is indivisible (why it's important to avoid multiphoton events)

p_{Eve} = probability that Eve guesses any given bit correctly

$p_{Eve} = \left(\frac{1}{2} + \varepsilon\right)$ = ε - security definition
↓
small number

110 | 010 | 110 | 110 | 001

0 1 0 0 1

Ekert 91 (PRL) - uses entanglement

① Alice prepares EPR ↓ $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ → send qubit 2 → Bob

② Alice & Bob choose randomly between +, ×

③ Once this has been repeated many times, reveal the choices publicly

④ Where they agree, use as key
" " disagree use Bell inequality

Show no-signalling theorem — Ghirardi, Rimini, Weber 1980

i.e. the choice of Alice's measurements, does not change the probability of Bob's measurement outcomes. (P_B)

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)$$

p^{zz}_{AB} — choices; AB — to denote joint prob.

$$p^{zz}_{AB}(0,0) = 0 = p^{zz}_{AB}(1,1)$$

$$p^{zz}_{AB}(0,1) = \frac{1}{2} = p^{zz}_{AB}(1,0)$$

$$p^{z}_{B}(0) = p^{zz}_{AB}(0,0) + p^{zz}_{AB}(1,0) = \frac{1}{2}$$

$$p^{z}_{B}(1) = p^{zz}_{AB}(0,1) + p^{zz}_{AB}(1,1) = \frac{1}{2}$$

$A\{+|\psi^-\rangle$
$A\{\leftrightarrow|\psi^-\rangle$

If Alice measures on X, Bob on Z

$$p^{xz}_{AB}(0,0) = \frac{1}{4} = p^{xz}_{AB}(0,1)$$

$$p^{xz}_{AB}(1,0) = \frac{1}{4} = p^{xz}_{AB}(1,1)$$

$$p^{z}_{B}(0) = \frac{1}{2}$$

$$p^{z}_{B}(1) = \frac{1}{2}$$